

## CLASS TOPIC

# Simple linear regression and residual diagnostics

## What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

**W12****Statistics**

Clean formulas · aligned layout · corrected te



## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Economic question first

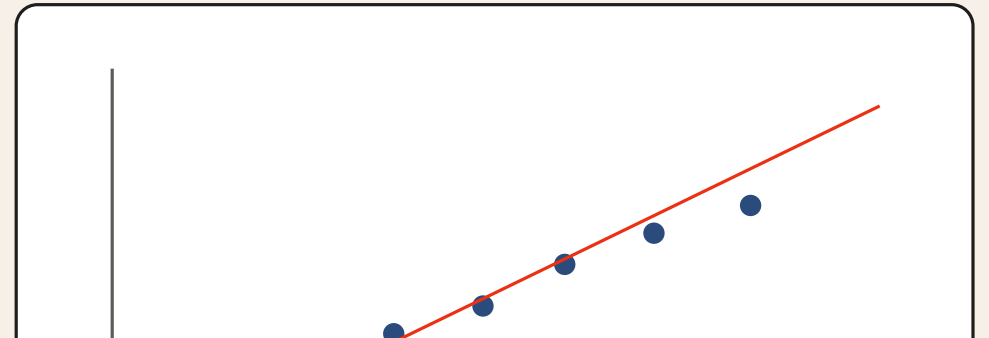
- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



## CLASS TOPIC

# Simple linear regression and residual diagnostics

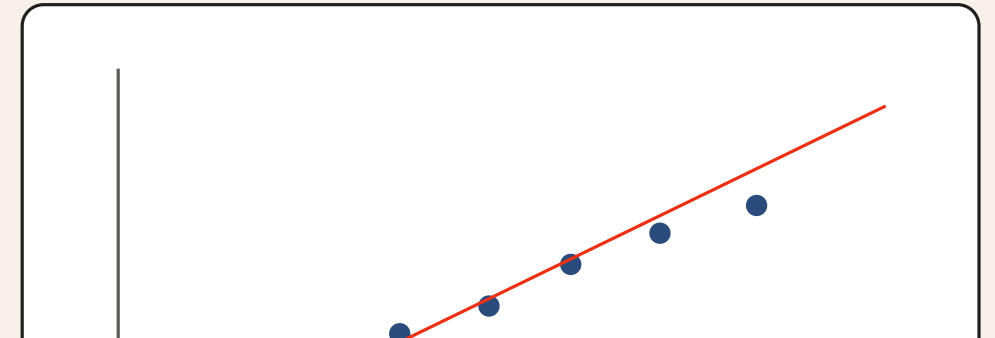
## Key definitions

- Population: full group of interest.
- Sample: observed part of the population.
- Statistic: number computed from a sample;  
parameter: number for the population.

## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$



## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Main formula box

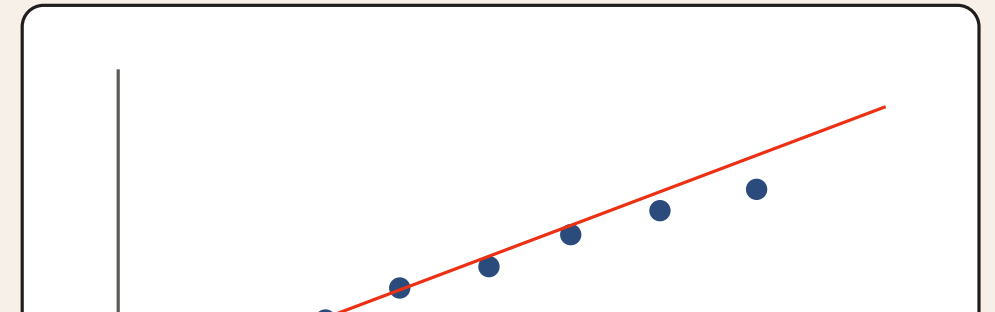
- All formulas are kept in one merged formula area on this slide.
- Read every symbol before calculating.
- Use parentheses carefully.

## FORMULA 1

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Mini example

- Write the data clearly.
- Substitute values step by step.
- Round only at the final reporting stage.

**01** Data: 4, 7, 8, 9, 12

**02** Step 1: compute the needed statistic

**03** Step 2: write the result with units.

**04** Step 3: explain it in one sentence.

**FORMULA**

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

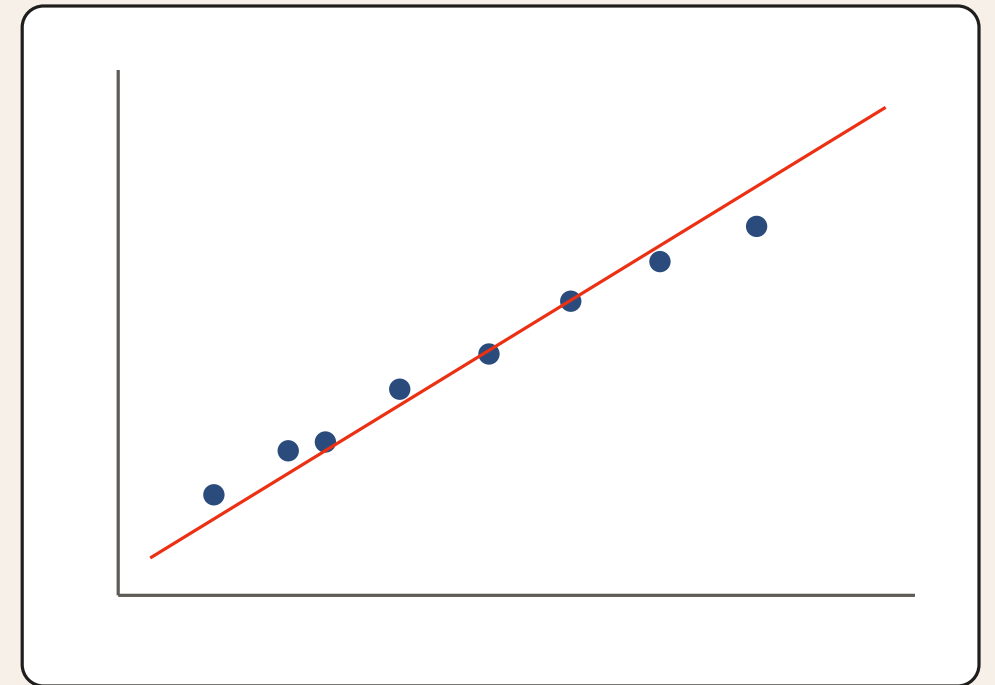
$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$

## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Visual intuition

- A graph should answer a question, not decorate the slide.
- Always label axes and units.
- Look for center, spread, shape and unusual values.



## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Second formula box

- This formula is separated from explanations but kept in one clean box.
- No broken equation fragments or floating text boxes.
- Notation is consistent across English and Russian versions.

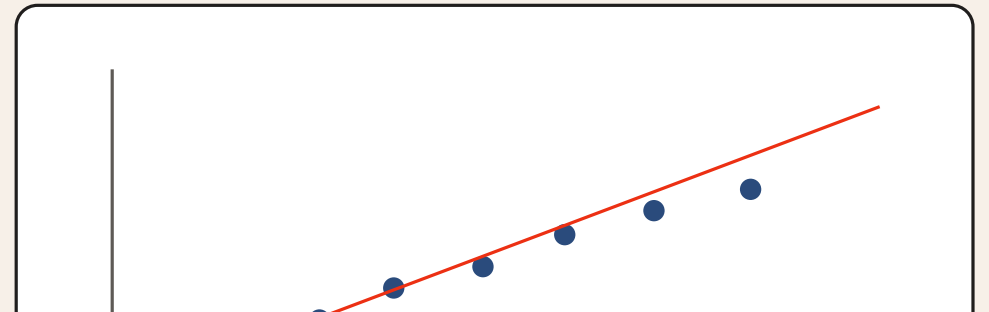
**FORMULA 2**

$$e_i = y_i - \hat{y}_i$$

$$SSE = \sum e_i^2$$

OLS chooses  $b_0, b_1$  to minimize SSE

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



## CLASS TOPIC

# Simple linear regression and residual diagnostics

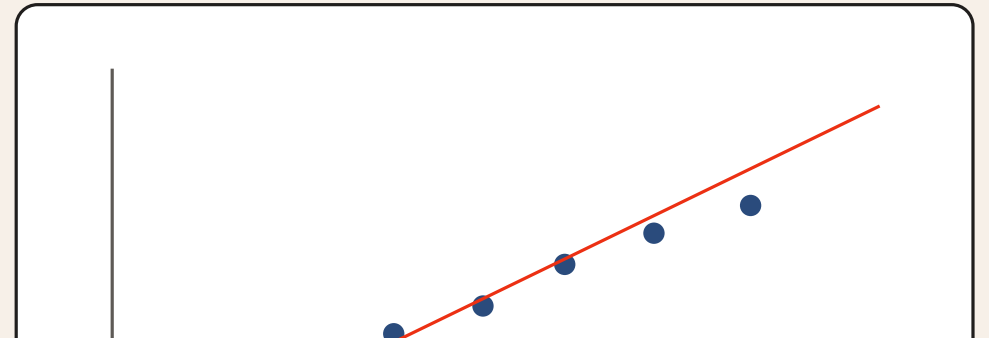
## Common mistakes

- Mixing sample and population symbols.
- Using a formula without checking conditions.
- Reporting a number without units or interpretation.

## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$



## CLASS TOPIC

# Simple linear regression and residual diagnostics

## R / spreadsheet check

- Use software to calculate, but keep the statistical logic visible.
- Check variable names, missing values and units first.
- Reproducible work is easier to grade and correct.

**R / EXCEL**

```
# example pattern  
mean(x); sd(x); summary(x)  
plot(x, y); cor(x, y)
```

**FORMULA**

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

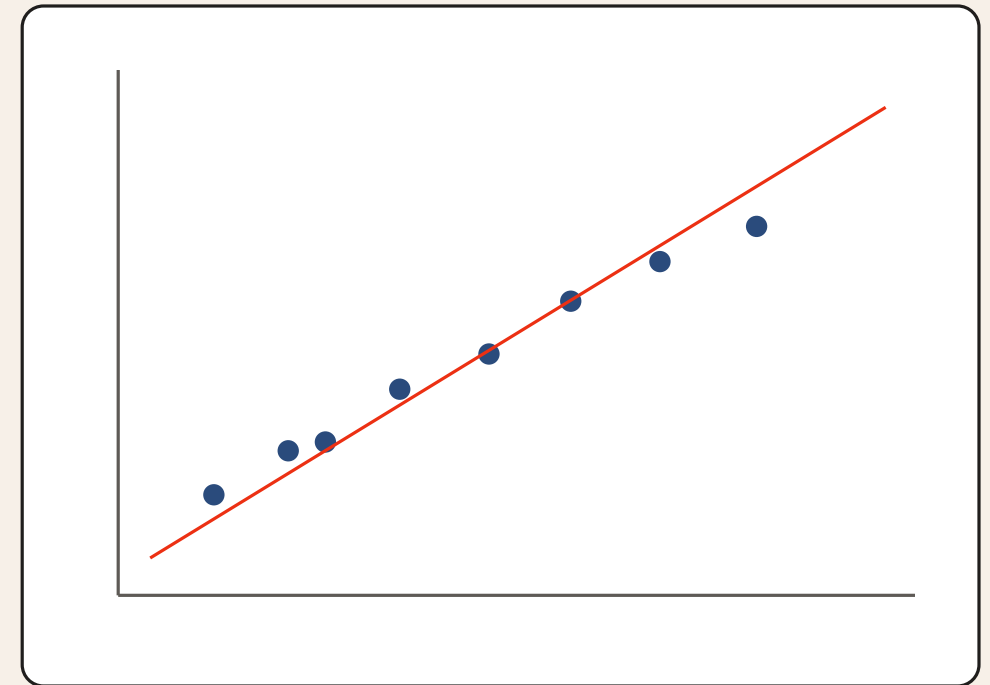
Estimated line:  $\hat{y}_i = b_0 + b_1 x_i$

## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.



## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

## CLASS TOPIC

# Simple linear regression and residual diagnostics

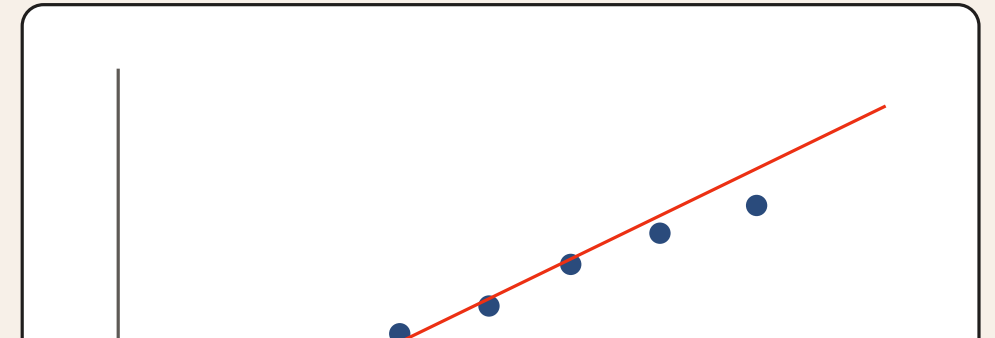
## Conditions and assumptions

- Check whether the method fits the data type.
- Check independence and sample size when needed.
- If assumptions fail, explain the limitation.

## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$



## CLASS TOPIC

# Simple linear regression and residual diagnostics

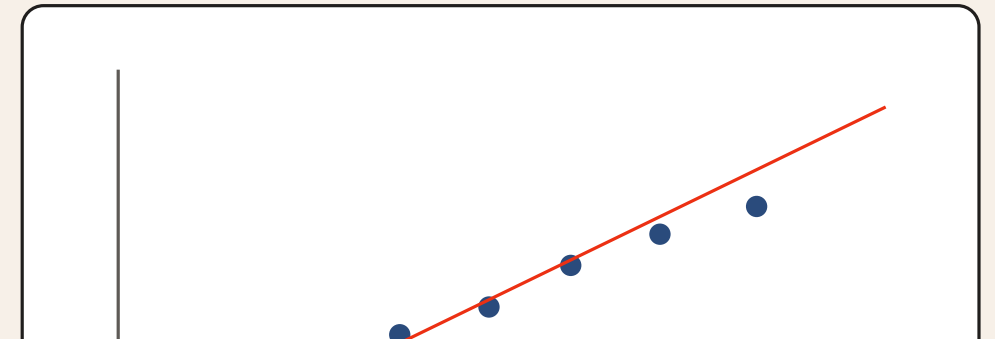
## Class exercise

- Solve one calculation by hand.
- Then verify it with software.
- Write a two-sentence conclusion.

## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$

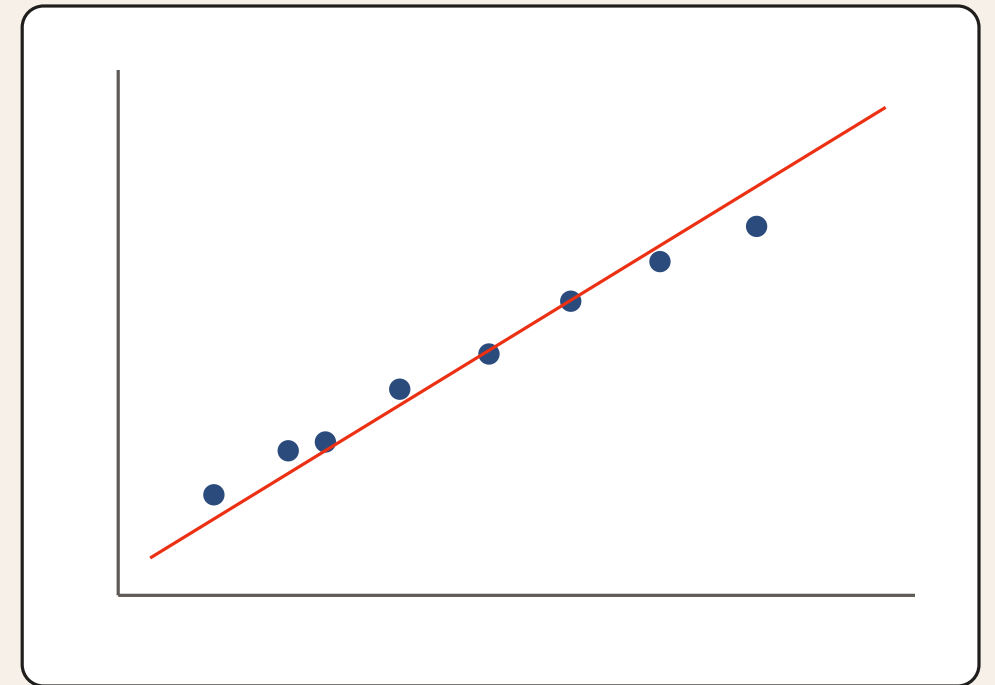


## CLASS TOPIC

# Simple linear regression and residual diagnostics

## How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
- Separate statistical evidence from causal claims.



## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

## CLASS TOPIC

# Simple linear regression and residual diagnostics

## Final checklist

- Question is clear.
- Variables and units are defined.
- Formula is correct.
- Result is interpreted.
- Limitations are stated.

## FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$

